

Assignment – 2

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College Name: IGDTUW

1. Function to print first 10 natural numbers

▶ # Function to print first 10 natural numbers

```
def print_numbers():  
    for i in range(1, 11):  
        print(i)
```

```
print_numbers()
```

```
... 1  
    2  
    3  
    4  
    5  
    6  
    7  
    8  
    9  
    10
```

2. Function to Calculate Sum of First N Natural Numbers

▶ # Function to calculate sum of first N natural numbers

```
def sum_n(n):  
    total = 0  
    for i in range(1, n + 1):  
        total += i  
    return total
```

```
n = int(input("Enter N: "))  
print("Sum =", sum_n(n))
```

```
... Enter N: 7  
    Sum = 28
```

3. Function to Reverse a Number

```
▶ # Function to reverse a number

def reverse_number(num):
    reverse = 0

    while num > 0:
        digit = num % 10
        reverse = reverse * 10 + digit
        num = num // 10

    return reverse

num = int(input("Enter a number: "))
print("Reversed number =", reverse_number(num))
```

```
... Enter a number: 45
Reversed number = 54
```

4. Function to Count Digits in a Number

```
▶ # Function to count digits

def count_digits(num):
    count = 0

    while num > 0:
        num = num // 10
        count += 1

    return count

num = int(input("Enter a number: "))
print("Number of digits =", count_digits(num))
```

```
... Enter a number: 4356
Number of digits = 4
```

5. Function to Check Palindrome Number

▶ # Function to check palindrome

```
def palindrome(num):
    temp = num
    rev = 0

    while num > 0:
        digit = num % 10
        rev = rev * 10 + digit
        num = num // 10

    if temp == rev:
        print("Palindrome Number")
    else:
        print("Not a Palindrome Number")

num = int(input("Enter a number: "))
palindrome(num)
```

```
... Enter a number: 454
    Palindrome Number
```

6. Function to Generate Fibonacci Series

▶ # Function to generate Fibonacci series

```
def fibonacci(n):
    a = 0
    b = 1

    for i in range(n):
        print(a, end=" ")
        c = a + b
        a = b
        b = c

n = int(input("Enter number of terms: "))
fibonacci(n)
```

```
... Enter number of terms: 4
    0 1 1 2
```

7. Calculator Using Functions

```
▶ def add(a, b):  
    return a + b  
def subtract(a, b):  
    return a - b  
def multiply(a, b):  
    return a * b  
def divide(a, b):  
    return a / b  
num1 = float(input("Enter first number: "))  
num2 = float(input("Enter second number: "))  
print("1. Addition")  
print("2. Subtraction")  
print("3. Multiplication")  
print("4. Division")  
choice = int(input("Enter choice: "))  
if choice == 1:  
    print("Result =", add(num1, num2))  
elif choice == 2:  
    print("Result =", subtract(num1, num2))  
elif choice == 3:  
    print("Result =", multiply(num1, num2))  
elif choice == 4:  
    if num2 != 0:  
        print("Result =", divide(num1, num2))  
    else:  
        print("Division by zero is not possible")  
else:  
    print("Invalid Choice")
```

```
... Enter first number: 23  
Enter second number: 48  
1. Addition  
2. Subtraction  
3. Multiplication  
4. Division  
Enter choice: 2  
Result = -25.0
```

8. Create a Text File and Store Student Details

```
 # Writing data into a text file

name = input("Enter student name: ")
roll = input("Enter roll number: ")

file = open("student.txt", "w")

file.write("Name : " + name + "\n")
file.write("Roll Number : " + roll)

file.close()

print("Data stored successfully.")
```

```
... Enter student name: yashi
Enter roll number: 233
Data stored successfully.
```

9. Read Data from a File

```
 # Reading data from a text file

file = open("student.txt", "r")

content = file.read()

print(content)

file.close()
```

```
... Name : yashi
Roll Number : 233
```

10. Handle Division by Zero Using Exception Handling

```
▶ # Exception handling example

try:
    num1 = int(input("Enter numerator: "))
    num2 = int(input("Enter denominator: "))

    result = num1 / num2

    print("Result =", result)

except ZeroDivisionError:
    print("Cannot divide by zero.")
```

```
... Enter numerator: 45
    Enter denominator: 4
all output actions = 11.25
```

11. Create a Student Class with Name and Marks

```
▶ # Student class

class Student:

    def __init__(self, name, marks):
        self.name = name
        self.marks = marks

    def display(self):
        print("Student Name :", self.name)
        print("Marks :", self.marks)

name = input("Enter student name: ")
marks = float(input("Enter marks: "))

s1 = Student(name, marks)

s1.display()
```

```
... Enter student name: Tanay
    Enter marks: 95
    Student Name : Tanay
    Marks : 95.0
```